

Advanced (Habanero)

- Q1.** A 12%—solution of acid is mixed with a 24%—solution of acid to get 160 liters of a 18%—solution. How many liters of the 12%—solution are used?
(A) 40 liters
(B) 80 liters
(C) 60 liters
(D) 100 liters
(E) 120 liters
- Q2.** A chemist mixes x liters of a 20%—salt solution with y liters of a 50%—salt solution. If the resulting mixture has a concentration of 30%—salt, which equation represents this situation?
(A) $0.2x + 0.5y = 0.3(x + y)$
(B) $0.2x - 0.5y = 0.3(x - y)$
(C) $0.2x + 0.5y = 0.3x + 0.3y$
(D) $0.2x - 0.5y = 0.3x - 0.3y$
(E) $0.2x + 0.5y = 0.6(x + y)$
- Q3.** A 30%—alcohol solution is mixed with a 70%—alcohol solution to create 20 liters of a 50%—alcohol solution. How many liters of the 30%—solution are used?
(A) 5 liters
(B) 10 liters
(C) 4 liters
(D) 6 liters
(E) 8 liters
- Q4.** A merchant has two types of coffee beans. Type A costs
5
per pound and Type B costs
3
per pound. How many pounds of Type B should be mixed with 5 pounds of Type A to get a mixture costing
4
per pound?
(A) 5 pounds
- (B) 3 pounds
(C) 2 pounds
(D) 3.5 pounds
(E) 4 pounds
- Q5.** A 10%—acid solution is mixed with x liters of a 30%—acid solution to get 12 liters of a 20%—acid solution. What is x ?
(A) 6 liters
(B) 8 liters
(C) 4 liters
(D) 10 liters
(E) 2 liters
- Q6.** A solution of 15%—salt water is mixed with 4 liters of a 25%—salt water solution to create 10 liters of a 20%—salt water solution. How many liters of the 15%—solution are used?
(A) 6 liters
(B) 4 liters
(C) 3 liters
(D) 2 liters
(E) 8 liters
- Q7.** What is the concentration of salt in the final mixture if 2 liters of a 30%—salt solution and x liters of a 10%—salt solution are mixed to create 6 liters of the final mixture?
(A) 16%
(B) 18%
(C) 20%
(D) 15%
(E) 12%
- Q8.** A 40%—alcohol solution is mixed with a 20%—alcohol solution to create 15 liters of a 30%—alcohol solution. How many liters of the 20%—solution are used?
(A) 10 liters
(B) 5 liters
(C) 3 liters
(D) 9 liters
(E) 6 liters

Open Ended Questions (FRQ)

- Q9.** A 25%—salt water solution is mixed with a 40%—salt water solution. If the resulting mixture is 30%—salt water and has a volume of 12 liters, determine the volumes of the two solutions used in the mixture.
- Q10.** A grocer mixes two types of nuts. Type A costs
8
per pound and Type B costs
12
per pound. If the grocer wants to create a 20-pound mixture costing
10
per pound, and Type A is used 2 times as much as Type B, find the amount of Type A and Type B used.

Chef's Tips

- Ensure students have all necessary materials before starting the test.
- Remind students to show all work and justify their answers where applicable.
- Make sure students understand that calculators are not allowed for this test.